PULLBACK OF THE UNIT GROUP SHEAF AND INTRINSICNESS OF THE LOCALLY RINGED CONDITION FOR MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains commutative diagrams typeset with Xy-pic; the retained two-slice preamble of this source supplies the native Xy-pic setup, so they compile as in the source.

Allowed standard prerequisites (editorial).

- Exactness of the pullback functor on sheaves. Stacks Project, Sites, section on morphisms of topoi (exactness of f^{-1}). Preserves the cartesianity of the diagram defining the units.
- Definition of a morphism of locally ringed topoi. Stacks Project, Modules on Sites, section on locally ringed topoi (the defining diagram). Supplies the condition verified in the second half of the lemma.

Source reference. Original lemma 04H7: sources/stacks-chow/tags.txt, source lines 5858–5858. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : Sh(\mathcal{C}') \to Sh(\mathcal{C})$ be a morphism of topoi. If $\mathcal{O}$ is a sheaf of rings on $\mathcal{C}$, then*

$$f^{-1}(\mathcal{O}^*) = (f^{-1}\mathcal{O})^*.$$

In particular, if $\mathcal{O}$ turns $\mathcal{C}$ into a locally ringed site, then setting $f^\sharp = id$ the morphism of ringed topoi

$$(f, f^\sharp) : (Sh(\mathcal{C}'), f^{-1}\mathcal{O}) \to (Sh(\mathcal{C}, \mathcal{O})$$

is a morphism of locally ringed topoi.

Proof. Note that the diagram

$$\begin{array}{ccc} \mathcal{O}^* & \longrightarrow & * \\ {\scriptstyle u\mapsto(u,u^{-1})}\downarrow & & \downarrow{\scriptstyle 1} \\ \mathcal{O}\times\mathcal{O} & \xrightarrow{(a,b)\mapsto ab} & \mathcal{O} \end{array}$$

is cartesian. Since f^{-1} is exact we conclude that

$$\begin{array}{ccc} f^{-1}(\mathcal{O}^*) & \longrightarrow & * \\ {\scriptstyle u\mapsto(u,u^{-1})}\downarrow & & \downarrow{\scriptstyle 1} \\ f^{-1}\mathcal{O}\times f^{-1}\mathcal{O} & \xrightarrow{(a,b)\mapsto ab} & f^{-1}\mathcal{O} \end{array}$$

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is cartesian which implies the first assertion. For the second, note that $(\mathcal{C}', f^{-1}\mathcal{O})$ is a locally ringed site by Lemma 04H7 so that the assertion makes sense. Now the first part implies that the morphism is a morphism of locally ringed topoi. $\square$